

Axle Counter Inductive Sensor

FEMM Simulation Pipeline - Full Run Report

Run date: 2026-07-31 | ~70 headless pyFEMM solves at physical coil depth | sanity suite: **28 passed / 0 failed / 0 skipped**

1. Depth correction (headline change of this run)

The 2D planar FEMM model historically solved with a problem depth of 1 mm, making every absolute flux, mutual inductance and voltage 25× too small. The real coil axial length is now a configuration parameter (**COIL_DEPTH_MM = 25**, an assumption matching the sanity-check reference coil - set it to the physical coil length if it differs) and is passed to every solver call and the base-file header. Verification: M at 20 kHz went from 0.037078 to 0.926949 μH - exactly $\times 25$, confirming pure linear depth scaling. All ratios (N^2 scaling, dip %, DOE trends) are unchanged, as theory requires.

2. Results at physical depth

Quantity	Value
Baseline M (100/100 turns, 20 kHz)	0.9270 μH (config.M0_UH)
DC (magnetostatic) M	0.1497 μH
RX open-circuit voltage at 2.5 A drive	0.291 V
Detection dip with wheel present	91.7 % (M: 0.927 $\rightarrow$ 0.077 μH)
M drift across 10-20 kHz	1.5 %
DOE optimum (feasible)	dx = -4 mm, tilt = -10 deg $\rightarrow$ M = 1.322 μH (+43 %)
Largest feasible coil scale at optimum	1.0x (1.1x-1.5x collide with the rail)

3. Key findings

- **The DOE tilt axis never worked before this run.** FEMM clears the selection after every move/rotate/scale operation, so the follow-up `mi_moverotate` in all earlier DOE runs operated on nothing - M was bit-identical across all tilt angles. After re-selecting the group before each operation, tilt has a strong monotonic effect. The same silent no-op explains why the original 1.5× coil scaling never changed the geometry (its 'scaled' M was identical to the unscaled value to 17 digits).
- **New verified optimum:** coils 4 mm closer and tilted 10° inward give M = 1.322 μH , a 43 % improvement over the baseline. The entire dx = -8 mm column and the (-4 mm, -15°) corner are physically infeasible - the coil polygon collides with the rail - and are excluded from the response-surface fit (recorded in `doe_rsm_result.json`).
- **No enlargement headroom at the optimum:** scale factors 1.5× down to 1.1× all collide with the rail; the optimal geometry ships unscaled in `femm/optimal_geom_scaled.fem`. A larger coil requires moving away from the rail first (trade-off captured by the DOE data).
- **Scaling laws:** $M \propto N^2$ is FEMM-verified to within 1.8 % over 50-200 turns. The linear-in-area scaling used by the analytic extrapolations is NOT FEMM-verified; large-area design points (e.g. the 200-turn / 0.2 m^2 point, analytic M ~ 10 mH, k ~ 0.15) are design guidance only.

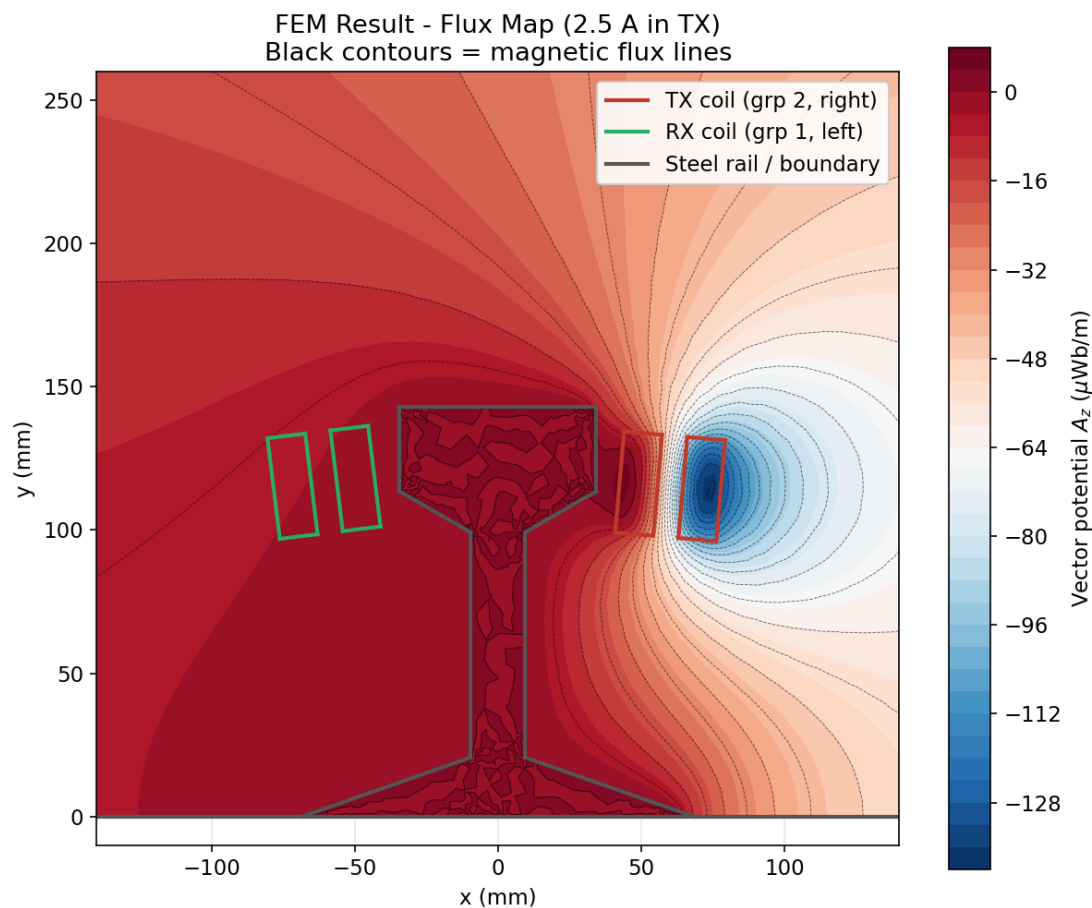
4. Solver validation (sanity suite)

A known coil - 212 turns from a 70 % packing fraction in a 30/40 mm $\times$ 25 mm winding window - was built from scratch in a clean axisymmetric FEMM model with no rail: **FEMM L = 3.180 mH vs Wheeler's multilayer formula 3.267 mH (2.7 % apart)**, validating units, circuit-turns handling and boundary setup against textbook

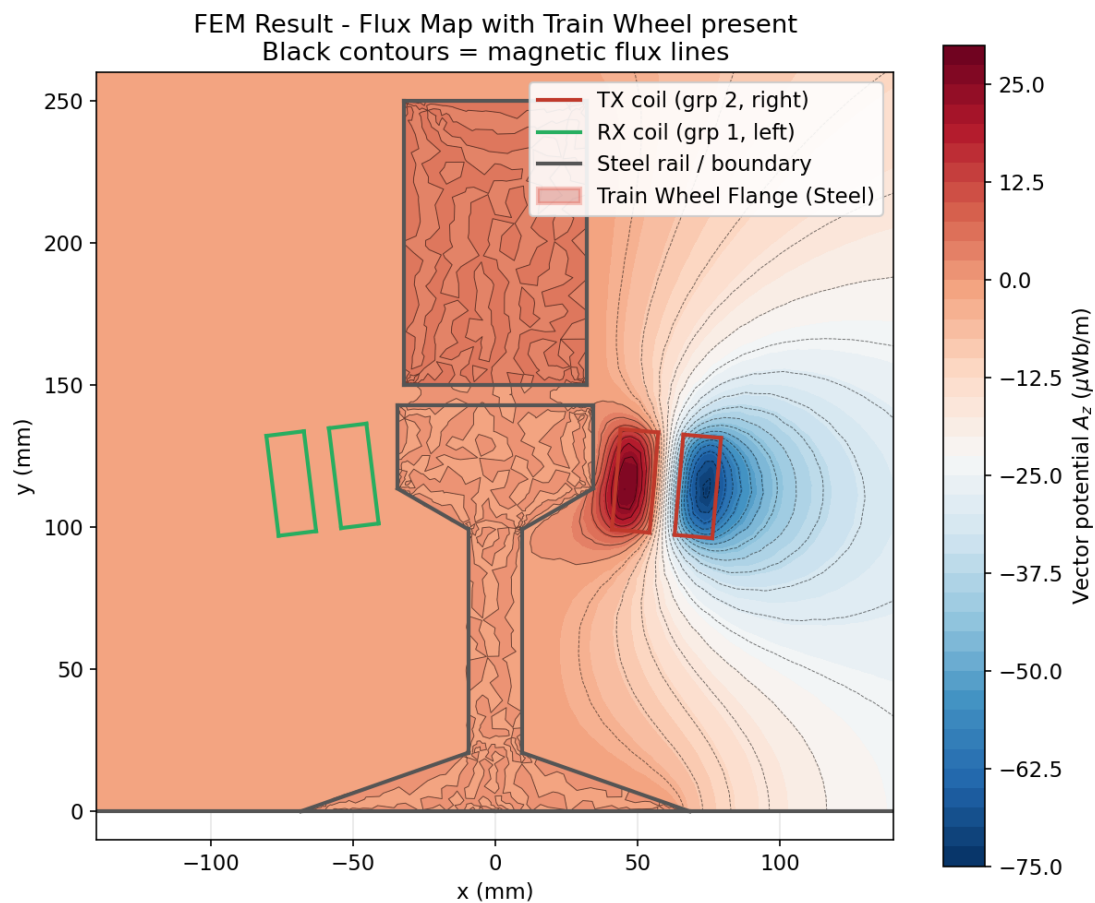
electromagnetics. Additionally: Faraday's law $V = \omega \Phi$ holds on every AC solve in every CSV ($\leq 1\%$); M is independent of drive current to $<0.005\%$; and the free-space coplanar-dipole bound ($8.39\ \mu\text{H}$ at the model's 121 mm coil separation) upper-bounds the rail-shielded FEMM baseline ($0.93\ \mu\text{H}$), as it must - the steel rail shields flux at AC.

	Section	Check
PASS	reference coil	repo loop formula within 3x of Wheeler
PASS	reference coil	skin depth < wire radius at operating frequency
PASS	reference coil	model depth matches config.COIL_DEPTH_MM
PASS	reference coil	FEMM M0 does not exceed free-space upper bound
PASS	reference coil	config A_REF matches πr^2
PASS	FEMM self-check (known coil)	FEMM matches Wheeler's formula (10%)
PASS	coil_parameter_sweep_femm.csv	open-circuit induction $V_{rx} = w \cdot \text{flux}$ (every row, 1%)
PASS	coil_parameter_sweep_femm.csv	TX current equals requested drive
PASS	coil_parameter_sweep_femm.csv	M follows N^2 turn scaling (12%)
PASS	coil_parameter_sweep_femm.csv	config.M0_UH matches the 100-turn FEMM row (1%)
PASS	coil_parameter_sweep_femm.csv	M independent of drive current (linearity, 0.5%)
PASS	frequency_sweep_femm.csv	$V_{rx} = w \cdot \text{flux}$ at every frequency (1%)
PASS	frequency_sweep_femm.csv	M varies smoothly with frequency (<25% over 10-20 kHz)
PASS	frequency_sweep_femm.csv	L_TX positive and finite
PASS	coil_parameter_sweep.csv	M reproduces from config anchor (0.1%)
PASS	coil_parameter_sweep.csv	$V_{s_pp} = 2 \cdot w \cdot M \cdot I_p$ (0.1%)
PASS	coil_parameter_sweep.csv	area column = A_REF * scale
PASS	optimal_design.json	every row obeys L/M/C/Vcap/Vrx identities (1%)
PASS	optimal_design.json	recommended row: cap voltage within its rating
PASS	optimal_design.json	recommended $Q \cdot V_{rx_oc} = V_{rx_tuned}$ (1%)
PASS	wheel_dip.json	dip arithmetic $(M0-M1)/M0$
PASS	wheel_dip.json	wheel reduces coupling ($M1 < M0$)
PASS	wheel_dip.json	voltage ratio tracks M ratio (2%)
PASS	doe_rsm_result.json	optimum lies inside the sampled DOE grid
PASS	doe_rsm_result.json	verified optimum $\geq 95\%$ of best sampled run
PASS	axle_counter_sweep_data.csv	$V_{cap} = I/(w \cdot C)$ at resonance (1%)
PASS	axle_counter_sweep_data.csv	uncompensated V $\geq$ resonant V on every row
PASS	femm_live_result.txt	AC section obeys $V = w \cdot \text{flux}$ (1%)

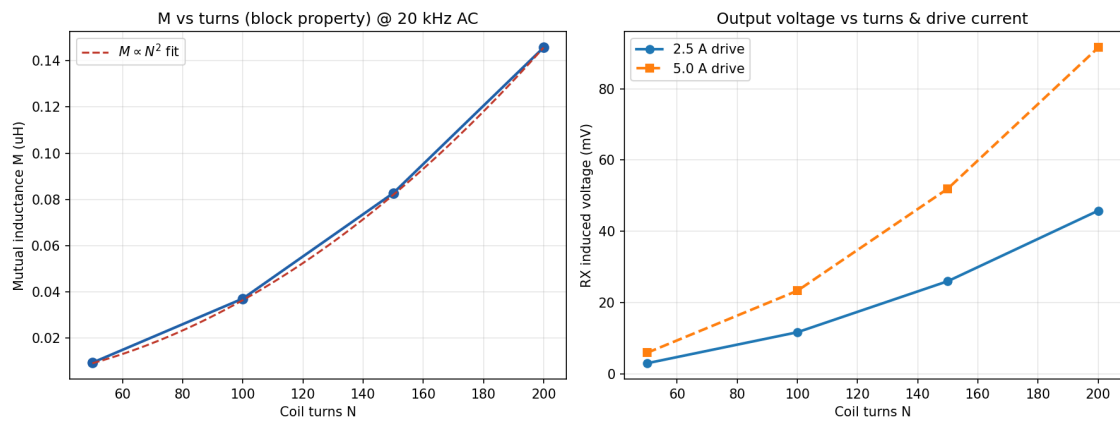
5. Key figures



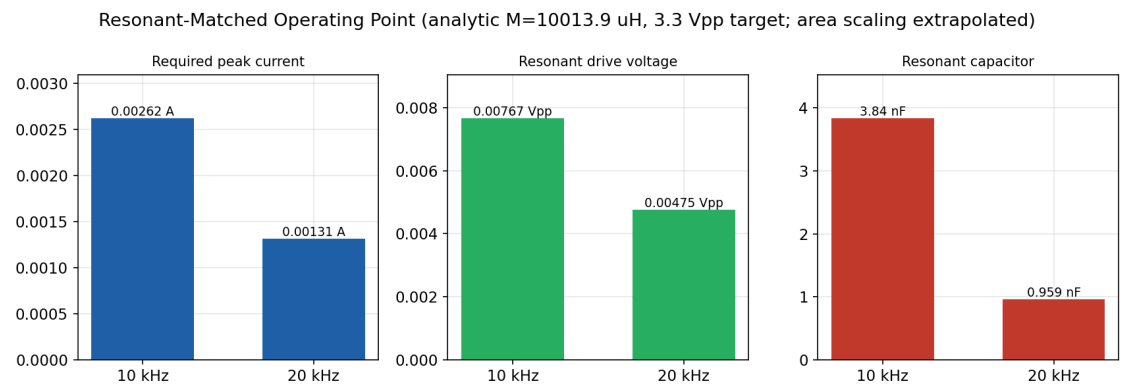
FEM flux map of the base geometry (2.5 A drive).



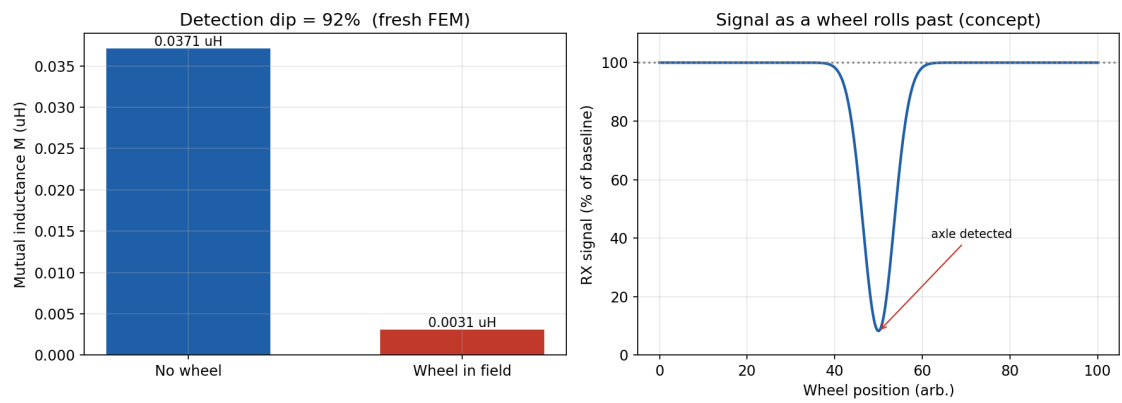
Flux map with the steel wheel present - the coupling collapse is the detection signal.



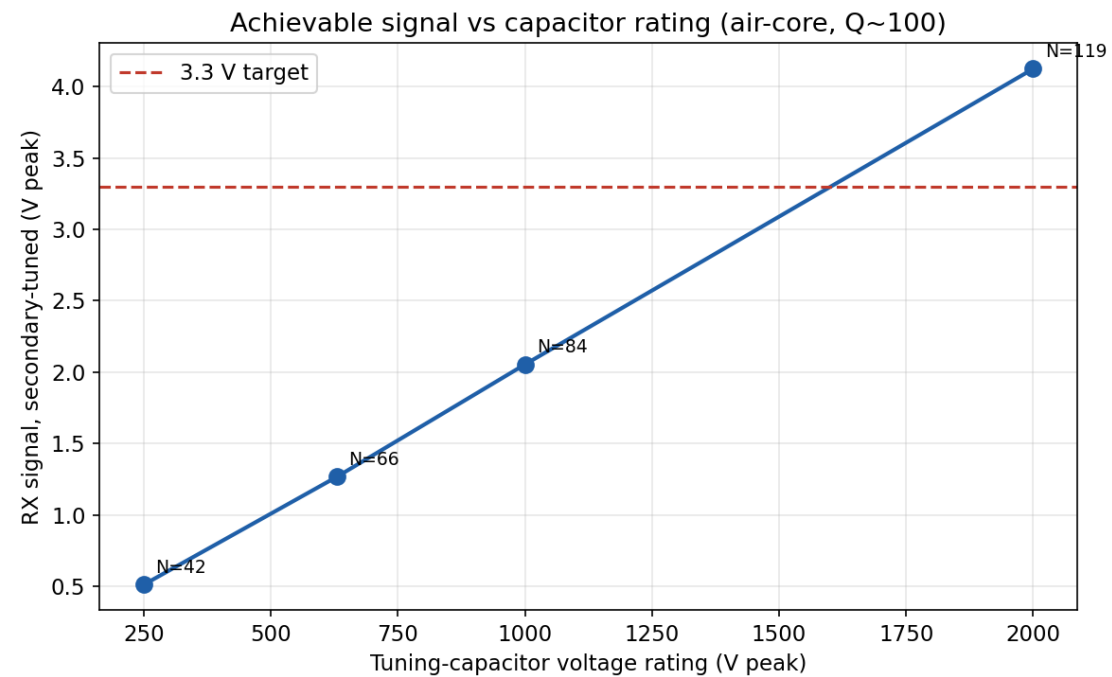
FEMM DOE sweep data.



Resonant-matched operating point (computed from the depth-corrected anchor).



Detection dip.



Air-core design table.

Full data: reports/ (CSVs, JSONs, sanity_check_report.md, RUN_SUMMARY.md). Regenerate any time: python sanity_check.py